

Figure 5: An instruction inserted at each stage in a pipelined processor

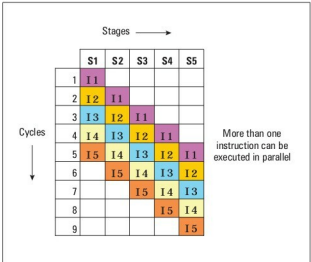


Figure 6: A pipelined processor

as an example. The stages in the RISC pipeline (shown in Figure 8) are:

1. Instruction fetch cycle (IF)
2. Instruction decode/register fetch cycle (ID)
3. Execution/Effective address cycle (EX)
4. Memory access (MEM)
5. Write-back cycle (WB)

Let's look at the meaning of each of these stages:

1. Instruction Fetch (IF): In this stage, the content of the Program Counter is used to access memory and fetch the next instruction to be executed.
2. Instruction Decode (ID): During this stage, the instruction is decoded, and the required operands are retrieved from the general-purpose registers or special-function registers.
3. Execute (EX): Calculations are performed during this stage. This includes effective address calculations for Load or Store instructions. The next Program Counter value is also calculated during this stage of the pipeline, so that branches, where applicable, can be executed. Some initial pre-calculation for memory decoding is also performed in this stage.
4. Memory Access (MA): If the instruction being executed is Load or Store, then the data memory is accessed during this stage. The previously calculated effective address is applied to the data memory, and the read or write is

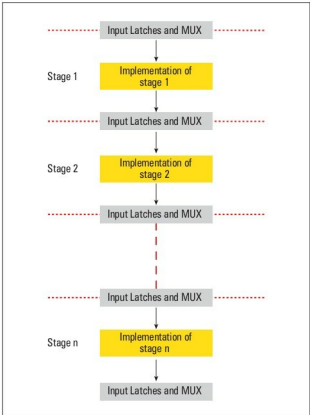


Figure 7: General structure of the pipeline

- performed in accordance with the instruction type.
5. Write-Back (WB): During this stage, the results of the calculation from the *Execute* stage, or the memory load from the Memory Access stage, are updated into the general-purpose registers or special-function registers.
- For example, let's suppose there are 5 instructions to be executed, and they appear in the order indicated in Table 1.

Address (PC)	Instruction
n	i0
n+1	i1
n+2	i2
n+3	i3
n+4	i4
n+5	i5

Table 1: Sample instruction set

These instructions will be executed in our pipeline as shown in Figure 9.

As we see in Figure 9, a standard instruction consumes five clock cycles, from an instruction fetch to write-back. However, since five instructions are concurrently processed by the pipeline, an instruction can be executed for every cycle.

Now, as the famous saying goes, all that glitters is